

AWS maintenance / establishment checklist

Station name	ZAK M
People present	NELI DANIEL BINDER (Instructed by:)
Purpose of visit	routine maintenance
Date & time	Arrival: 20/6/2015 12.00 Departure: 20/4/2015
Transport	By: snow scooter From: Locking Down
Weather	fair, then cloudy

Remarks on state of AWS / maintenance / establishment

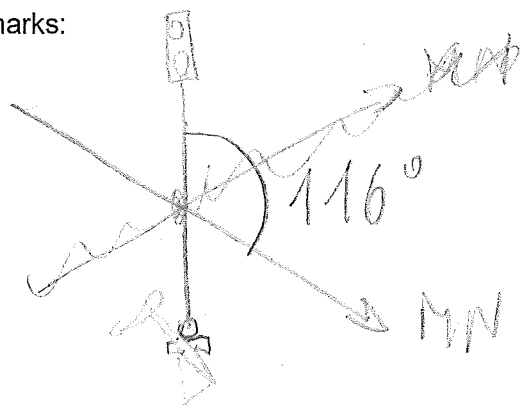
good outside, snow to mid-logger box level

Inside logger box there was some snow on the bottom right corner and some refrozen condensation/hoar on the Indian equipment in the top left corner. Dried out as feasible and put 6 desiccant bags inside. Would be good to put screws into keep the door closed. Which size are they?

Metadata upon arrival (Establishment of new AWS: Go to last page)

Time difference between logger clock and UTC	+ / -
Adjust logger time to UTC (be sure PC runs on UTC time)	☒ Y / ☐ N
Download data to PC and/or change CF Card (wait for green light)	☒ Y / ☐ N
Name of logger program	
Photo or screen dump of values in fast scan mode	Y / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Altitude (m)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	☒ Y / ☐ N
Mast tilt in boom direction (+ if radiometer tilting down)	-4.5 °
Mast tilt across boom looking from radiometer (- if clockwise)	-1.0 °
Boom direction relative to north	See sketch ° true/magn
Wind sensor box direction along boom? If not, measure direction.	☒ Y / ☐ N
Radiometer aligned with mast and inclinometer? If not, measure.	☒ Y / ☐ N
Temperature / humidity height (bottom of casing – surface)	156 cm
Sonic ranger on AWS height (membrane – surface)	164 cm
Wind sensor height (center of propeller – surface)	207 cm
Sonic ranger on stakes height (membrane – surface)	cm
Free length of stake – outer, holding boom with sonic ranger	cm
Free length of stake – middle	cm
Free length of stake – outer, holding boom without sonic ranger	cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	Buried cm
Height of ablation hose fluid level above the ice surface	165 cm
Vertical difference in surface height (at station – at hose) (approx)	BURIED cm
Length of thermistor string on surface from surface marking	BURIED cm
Height of surface irregularities	cm 2
Stake assembly still standing?	Y / ☒ N / more or less
Apparent damage to sensors? If yes, which?	Y / ☒ N

Remarks:



Vertical bottom of 1st tubing: 1.89 m

positive signal bottom first phase: 1.45 m

Metadata after maintenance / establishment

Time difference between logger clock and UTC	①	+ / -
Adjust logger time to UTC (be sure PC runs on UTC time)		(Y) / N
Download data to PC and/or change CF Card (wait for green light)		(Y) / N
Name of logger program		
Photo or screen dump of values in fast scan mode		(Y) / N
Latitude (dd mm.mmm)		
Longitude (dd mm.mmm)		
Altitude (m)		
Photos of tripod, sensors, logger box wiring etc. (more is better)		(Y) / N
Mast tilt in boom direction (+ if radiometer tilting down)	+0.5	°
Mast tilt across boom looking from radiometer (- if clockwise)	-4.5	°
Boom direction relative to north (radiometer should point to true south)		° true/magn
Wind sensor box direction along boom? If not, change.		Y / N
Radiometer aligned with mast and inclinometer? If not, align.		Y / N
Temperature / humidity height (bottom of casing – surface)	158	cm
Sonic ranger on AWS height (membrane – surface)	164	cm
Wind sensor height (center of propeller – surface)	208 208	cm
Sonic ranger on stakes height (membrane – surface)	/	cm
Free length of stake – outer, holding boom with sonic ranger	/	cm
Free length of stake – middle	/	cm
Free length of stake – outer, holding boom without sonic ranger	/	cm
Free length of ablation hose on/above the ice surface	9.60 m	cm
Alt: depth of ablation hose using markings on hose	No MARKINGS, HOLE: 16 m	cm
Height of ablation hose fluid level above the ice surface	163.12 m	cm
Vertical difference in surface height (at station – at hose) (approx)	0	cm
Length of thermistor string on surface from surface marking	-11	cm
Height of surface irregularities	snow 2	cm

Remarks:

NOTE: the boom was re-oriented to face ~~the~~ South better. Solar panel also re-oriented accordingly

Solar panel solidly on hole bottom.



64 58 41.87

Serial numbers and maintenance details

Col. SW 17.07 LWR 13.67
SW 10.79 LWR 12.23

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	110562	110562	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	SN750	26-750	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna	?	?	
Wind sensor	105090	=	
Temperature / humidity assembly (Fan, Bracket, MP100, Hygroclip)	ASS: 4/2012 FAN: 55210003 =	=	- New HygroClip with old assembly: Y/N - Fan spins and sounds OK: Y/N Note Ass no. + serialno. for all 4 parts!
Or: HygroClip only	MP100-60079558 = S3: 55025152 NEW	=	- Large offset in HygroClip temperature: Y/N
Sonic ranger on AWS	S30L2: 60163838	163838	- Old sensor with new membrane: Y/N - Clicking as it should: Y/N
Solar panel	3503 = C10702026	139250 =	- Output OK: Y/N - Clean: Y/N
GPS antenna	28670151		- Stuck to top of logger enclosure: Y/N
Data logger	E1350	=	- New internal battery in old logger: Y/N - New operating system: Y/N - New logger program: Y/N
Card reader	3355	=	- LED either green or flashing orange (active): Y/N
Multiplexer	E3903		- Clicking as it should: Y/N
Iridium modem IMEI number	300036012437350 =		- Transmissions tested: Y/N - Problems: Y/N
Barometer	3440409	=	
Sonic ranger on stakes			- Old sensor with new membrane: Y/N - Clicking as it should: Y/N
Thermistor string	B14	B13	- New string in use, but old one still in ice: Y/N
Ablation hose assembly	ABS 5003	No NUMBER?	- New hose in use, but old one still in ice: Y/N - Air in current hose: Y/N - Air removed from current hose: Y/N - Added 50% antifreeze mix to bladder: Y/N
Logger enclosure			- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N
Battery box			- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N
Stakes			- New ones drilled in: Y/N - Length: 2 / 4 / 6 / 8 / 10 m
Tripod			- New parts: Y/N

Hygroclip in positive shield: 60163836 (new)

60192374 (old)

Versale s/n: K059004

AWS maintenance / establishment checklist

Station name	ZAY - 100 S	
People present	Neil DAVIES	(Instructed by:)
Purpose of visit	ROUTINE MAINTENANCE	
Date & time	Arrival: 27.4.2015	Departure:
Transport	By: Shearwater	From: 20 Chesham Rd. St.
Weather	Sunny, little wind	

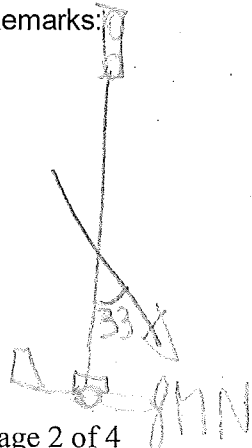
Remarks on state of AWS / maintenance / establishment

Solar panel for cwy & W

Metadata upon arrival (Establishment of new AWS: Go to last page)

Time difference between logger clock and UTC	+ / -
Adjust logger time to UTC (be sure PC runs on UTC time)	Y / N
Download data to PC and/or change CF Card (wait for green light)	Y / N
Name of logger program	
Photo or screen dump of values in fast scan mode	Y / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Altitude (m)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	0 °
Mast tilt across boom looking from radiometer (- if clockwise)	-2 °
Boom direction relative to north	see map ° true/magn
Wind sensor box direction along boom? If not, measure direction.	Y / N
Radiometer aligned with mast and inclinometer? If not, measure.	Y / N
Temperature / humidity height (bottom of casing – surface)	118 cm
Sonic ranger on AWS height (membrane – surface)	127 cm
Wind sensor height (center of propeller – surface)	170 cm
Sonic ranger on stakes height (membrane – surface)	111 cm
Free length of stake – outer, holding boom with sonic ranger	0.115 m cm
Free length of stake – middle	0.115 m cm
Free length of stake – outer, holding boom without sonic ranger	167 cm
Free length of ablation hose on/above the ice surface	2.28 m cm
Alt: depth of ablation hose using markings on hose	? cm
Height of ablation hose fluid level above the ice surface	? cm
Vertical difference in surface height (at station – at hose) (approx)	? cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	cm
Stake assembly still standing?	Y / N / more or less
Apparent damage to sensors? If yes, which?	Y / N

Remarks:



tilt meter angle across boom
is -3, radiometer is -2

Serial numbers and maintenance details

ZACK - S
col. 11.58 x 10 - 6 V/W m

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	071365	100428	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	SN00055	=	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna			
Wind sensor	SN124027	122925	6027 6761 6069 0864
Temperature / humidity assembly (Fan, Bracket, MP100, Hygroclip)	FAN: 6025 9495 HP100: ? S3: ?	ASS: 3/2011 RS12T HYCPC	- New HygroClip with old assembly: Y/N - Fan spins and sounds OK: Y/N Note Ass no. + serialno. for all 4 parts!
Or: HygroClip only		MP100: 6069 0862 S3: 6091 550	- Large offset in HygroClip temperature: Y/N
Sonic ranger on AWS		2840	- Old sensor with new membrane: Y/N - Clicking as it should: Y/N
Solar panel	OLD WRECK	=	- Output OK: Y/N - Clean: Y/N
GPS antenna			- Stuck to top of logger enclosure: Y/N
Data logger			- New internal battery in old logger: Y/N - New operating system: Y/N - New logger program: Y/N
Card reader			- LED either green or flashing orange (active): Y/N
Multiplexer			- Clicking as it should: Y/N
Iridium modem IMEI number			- Transmissions tested: Y/N - Problems: Y/N
Barometer			
Sonic ranger on stakes			- Old sensor with new membrane: Y/N - Clicking as it should: Y/N
Thermistor string			- New string in use, but old one still in ice: Y/N
Ablation hose assembly			- New hose in use, but old one still in ice: Y/N - Air in current hose: Y/N - Air removed from current hose: Y/N - Added 50% antifreeze mix to bladder: Y/N
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 2 / 4 / 6 / 8 / 10 m		
Tripod	- New parts: Y/N		

col's.

SWA 12.13
SWA 12.38
CWA 6.38
CWA 4.26

10-6

Metadata after maintenance / establishment

Time difference between logger clock and UTC	+ / -
Adjust logger time to UTC (be sure PC runs on UTC time)	Y / N
Download data to PC and/or change CF Card (wait for green light)	Y / N
Name of logger program	
Photo or screen dump of values in fast scan mode	Y / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Altitude (m)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	°
Mast tilt across boom looking from radiometer (- if clockwise)	°
Boom direction relative to north (radiometer should point to true south)	° true/magn
Wind sensor box direction along boom? If not, change.	Y / N
Radiometer aligned with mast and inclinometer? If not, align.	Y / N
Temperature / humidity height (bottom of casing – surface)	cm
Sonic ranger on AWS height (membrane – surface)	cm
Wind sensor height (center of propeller – surface)	cm
Sonic ranger on stakes height (membrane – surface)	cm
Free length of stake – outer, holding boom with sonic ranger	cm
Free length of stake – middle	cm
Free length of stake – outer, holding boom without sonic ranger	cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	cm
Vertical difference in surface height (at station – at hose) (approx)	cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	cm

Remarks:

AWS maintenance / establishment checklist

2015

Station name	ZAC - T...	
People present	ACIT, DANIEL BINDER (Instructed by:)	
Purpose of visit		
Date & time	Arrival: 26.4.13:10	Departure:
Transport	By: Snow	From: Zechenbergt.
Weather		

Remarks on state of AWS / maintenance / establishment

Almost entirely buried.
 Only wind monitor
 and Instruk Aut. (top of)
 sticking out of snow.

Metadata upon arrival (Establishment of new AWS: Go to last page)

Time difference between logger clock and UTC	+ / -
Adjust logger time to UTC (be sure PC runs on UTC time)	Y / N
Download data to PC and/or change CF Card (wait for green light)	Y / N
Name of logger program	
Photo or screen dump of values in fast scan mode	Y / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Altitude (m)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	+6.2°
Mast tilt across boom looking from radiometer (- if clockwise)	-2.7°
Boom direction relative to north	° true/magn
Wind sensor box direction along boom? If not, measure direction.	Y / N
Radiometer aligned with mast and inclinometer? If not, measure.	Y / N
Temperature / humidity height (bottom of casing – surface)	cm
Sonic ranger on AWS height (membrane – surface)	cm
Wind sensor height (center of propeller – surface)	38 cm
Sonic ranger on stakes height (membrane – surface)	cm
Free length of stake – outer, holding boom with sonic ranger	cm
Free length of stake – middle	cm
Free length of stake – outer, holding boom without sonic ranger	cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	cm
Vertical difference in surface height (at station – at hose) (approx)	cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	cm
Stake assembly still standing?	Y / N / more or less
Apparent damage to sensors? If yes, which?	Y / N

Remarks:

measured before digging

Serial numbers and maintenance details

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	080163		- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	00045		- Aligned with radiometer within 0.5°: Y/N
Satellite antenna	6015 13		
Wind sensor	82157		
Temperature / humidity assembly (Fan, Bracket, MP100, Hygroclip)	Ass 2/2011 MP100: 6069 0861 S3: 60163833	6033240 S3: 55029226	- New HygroClip with old assembly: Y/N - Fan spins and sounds OK: Y/N Note Ass no. + serialno. for all 4 parts !
Or: HygroClip only			- Large offset in HygroClip temperature: Y/N
Sonic ranger on AWS	SR 50A SN: 2878		- Old sensor with new membrane: Y/N - Clicking as it should: Y/N
Solar panel	501 6031		- Output OK: Y/N - Clean: Y/N
GPS antenna			- Stuck to top of logger enclosure: Y/N
Data logger			- New internal battery in old logger: Y/N - New operating system: Y/N - New logger program: Y/N
Card reader			- LED either green or flashing orange (active): Y/N
Multiplexer			- Clicking as it should: Y/N
Iridium modem IMEI number			- Transmissions tested: Y/N - Problems: Y/N
Barometer			
Sonic ranger on stakes			- Old sensor with new membrane: Y/N - Clicking as it should: Y/N
Thermistor string			- New string in use, but old one still in ice: Y/N
Ablation hose assembly			- New hose in use, but old one still in ice: Y/N - Air in current hose: Y/N - Air removed from current hose: Y/N - Added 50% antifreeze mix to bladder: Y/N
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N		
Stakes	- New ones drilled in: Y/N - Length: 2 / 4 / 6 / 8 / 10 m		
Tripod	- New parts: Y/N		

Metadata after maintenance / establishment

Time difference between logger clock and UTC	50 PIC. +/-
Adjust logger time to UTC (be sure PC runs on UTC time)	Y/N
Download data to PC and/or change CF Card (wait for green light)	Y/N
Name of logger program	=
Photo or screen dump of values in fast scan mode	Y/N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Altitude (m)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N
Mast tilt in boom direction (+ if radiometer tilting down)	-10.5 °
Mast tilt across boom looking from radiometer (- if clockwise)	+2.0 °
Boom direction relative to north (radiometer should point to true south)	° true/magn
Wind sensor box direction along boom? If not, change.	Y/N
Radiometer aligned with mast and inclinometer? If not, align.	Y/N
Temperature / humidity height (bottom of casing – surface)	285 cm
Sonic ranger on AWS height (membrane – surface)	293 cm
Wind sensor height (center of propeller – surface)	~ 330 cm
Sonic ranger on stakes height (membrane – surface)	/ cm
Free length of stake – outer, holding boom with sonic ranger	/ cm
Free length of stake – middle	/ cm
Free length of stake – outer, holding boom without sonic ranger	/ cm
Free length of ablation hose on/above the ice surface	/ cm
Alt: depth of ablation hose using markings on hose	/ cm
Height of ablation hose fluid level above the ice surface	/ cm
Vertical difference in surface height (at station – at hose) (approx)	/ cm
Length of thermistor string on surface from surface marking	/ cm
Height of surface irregularities	5-10 cm

Remarks:

